

Numerical Example

On

KNN

Height (cms)	Weight (kgs)	T Shirt Size
158	58	M
158	59	M
158	63	M
160	59	M
160	60	M
163	60	M
163	61	M
160	64	L
163	64	L
165	61	L
165	62	L
165	65	L
168	62	L
168	63	L

Query : A new customer Sara has height 161 cm and weight 61 kg.
What size of T Shirt she needs ?

So, our new data point is (161, 61)

First two steps

1. $K = 5$
2. Find the Euclidean distance between (161, 61) with all the points in dataset

Euclidean Distance between (161, 61) and (158, 58)

$$= \sqrt{(161 - 158)^2 + (61 - 58)^2} = 4.2$$

Select the nearest neighbors and assign the class or size of T shirt

Height (cms)	Weight (kgs)	T Shirt Size	Distance	Neighbors
158	58	M	4.2	
158	59	M	3.6	
158	63	M	3.6	
160	59	M	2.2	3
160	60	M	1.4	1
163	60	M	2.2	3
163	61	M	2.0	2
160	64	L	3.2	5
163	64	L	3.6	
165	61	L	4.0	
165	62	L	4.1	
165	65	L	5.7	
168	62	L	7.1	
168	63	L	7.3	

5 - Nearest Neighbors

Height (cms)	Weight (kgs)	T Shirt Size	Distance	Neighbors
158	58	M	4.2	
158	59	M	3.6	
158	63	M	3.6	6
160	59	M	2.2	3
160	60	M	1.4	1
163	60	M	2.2	3
163	61	M	2.0	2
160	64	L	3.2	5
163	64	L	3.6	6
165	61	L	4.0	
165	62	L	4.1	
165	65	L	5.7	
168	62	L	7.1	
168	63	L	7.3	

7 - Nearest Neighbors

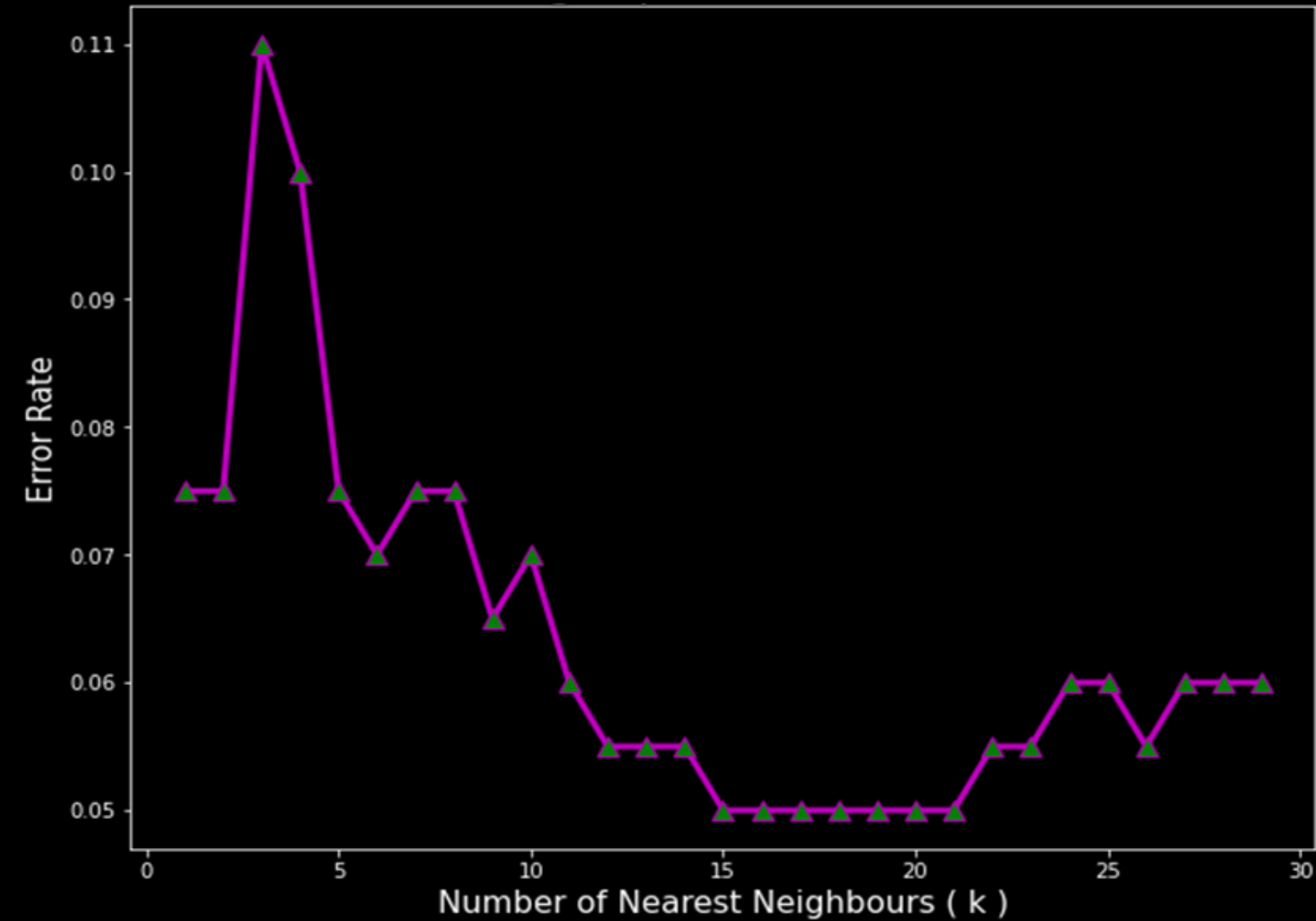


Height (cms)	Weight (kgs)	T Shirt Size	Distance	Neighbors
158	58	M	4.2	
158	59	M	3.6	8
158	63	M	3.6	6
160	59	M	2.2	3
160	60	M	1.4	1
163	60	M	2.2	3
163	61	M	2.0	2
160	64	L	3.2	5
163	64	L	3.6	6
165	61	L	4.0	9
165	62	L	4.1	
165	65	L	5.7	
168	62	L	7.1	
168	63	L	7.3	

9 - Nearest Neighbors

Finding Optimal Value of k

Finding Optimal Value of k



Error Rate = 1 - Accuracy

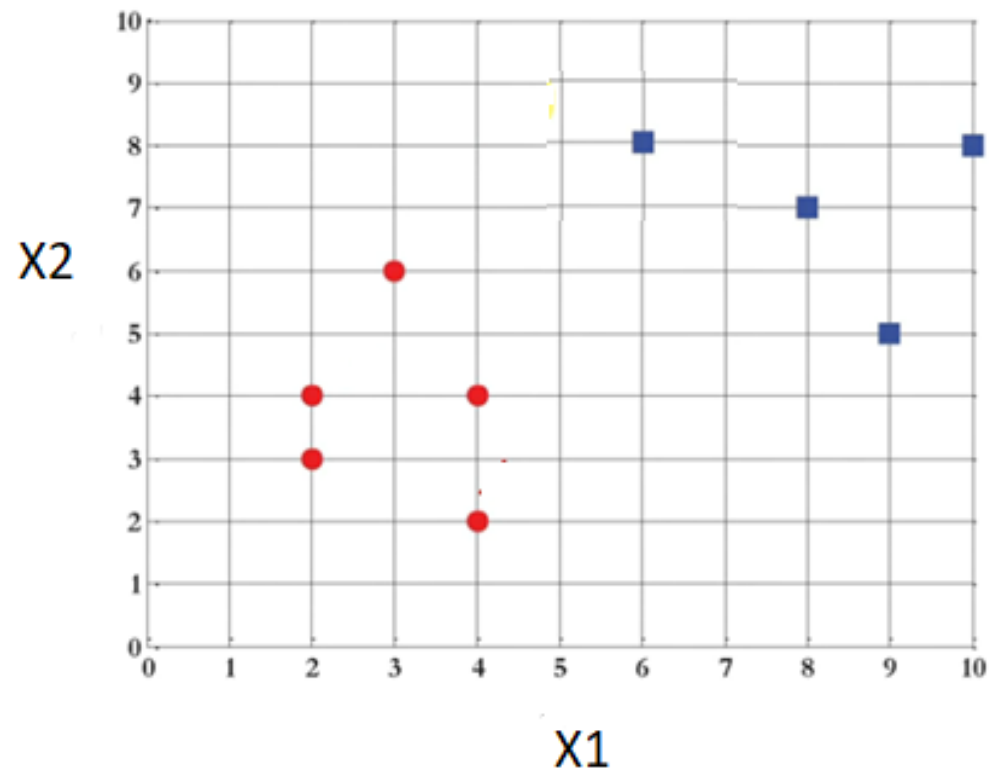
Numerical Example

On

LDA

Samples for class A : $\mathbf{X}_1 = (x_1, x_2) = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$

Sample for class B : $\mathbf{X}_2 = (x_1, x_2) = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$



Samples for class A : $\mathbf{X}_1=(x_1,x_2)=\{(4,2),(2,4),(2,3),(3,6),(4,4)\}$

Sample for class B : $\mathbf{X}_2=(x_1,x_2)=\{(9,10),(6,8),(9,5),(8,7),(10,8)\}$

The classes mean are :

$$\mu_1 = \frac{1}{N_1} \sum_{x \in \omega_1} x = \frac{1}{5} \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} 4 \\ 4 \end{pmatrix} \right] = \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}$$

$$\mu_2 = \frac{1}{N_2} \sum_{x \in \omega_2} x = \frac{1}{5} \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} + \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} 9 \\ 5 \end{pmatrix} + \begin{pmatrix} 8 \\ 7 \end{pmatrix} + \begin{pmatrix} 10 \\ 8 \end{pmatrix} \right] = \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix}$$

Covariance Matrix of class A

$$S1 = \sum \frac{(x - \mu_1)(x - \mu_1)^T}{N-1}$$

$$\begin{aligned} S1 &= \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right] \left[\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T + \left[\begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right] \left[\begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T \\ &+ \left[\begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right] \left[\begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T + \left[\begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right] \left[\begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T + \left[\begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right] \left[\begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \right]^T \bigg/_{N-1} \\ &= \begin{pmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{pmatrix} \end{aligned}$$

Similarly the covariance matrix of class B

$$S_2 = \begin{pmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{pmatrix}$$

Within-class scatter matrix:

$$\begin{aligned} S_w = S_1 + S_2 &= \begin{pmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{pmatrix} + \begin{pmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{pmatrix} \\ &= \begin{pmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{pmatrix} \end{aligned}$$

Between Class Scatter Matrix

$$\begin{aligned} S_B &= (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T \\ &= \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T \\ &= \begin{pmatrix} -5.4 \\ -3.8 \end{pmatrix} \begin{pmatrix} -5.4 & -3.8 \end{pmatrix} \\ &= \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{pmatrix} \end{aligned}$$

Calculate the Projection vector

$$\begin{aligned} w &= S_W^{-1}(\mu_1 - \mu_2) = \begin{pmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{pmatrix}^{-1} \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \\ &= \begin{pmatrix} 0.3045 & 0.0166 \\ 0.0166 & 0.1827 \end{pmatrix} \begin{pmatrix} -5.4 \\ -3.8 \end{pmatrix} \\ &= \begin{pmatrix} 0.9088 \\ 0.4173 \end{pmatrix} \end{aligned}$$

Calculating LDA

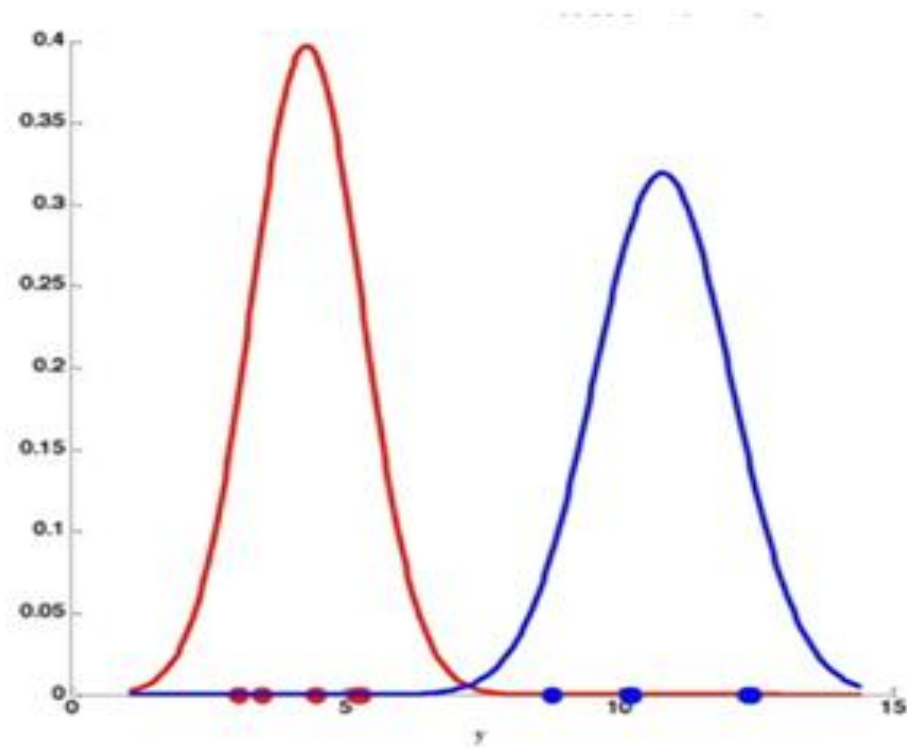
$$w = \begin{pmatrix} 0.9088 \\ 0.4173 \end{pmatrix}$$

Samples for class A : $\mathbf{X}_1 = (x_1, x_2) = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$

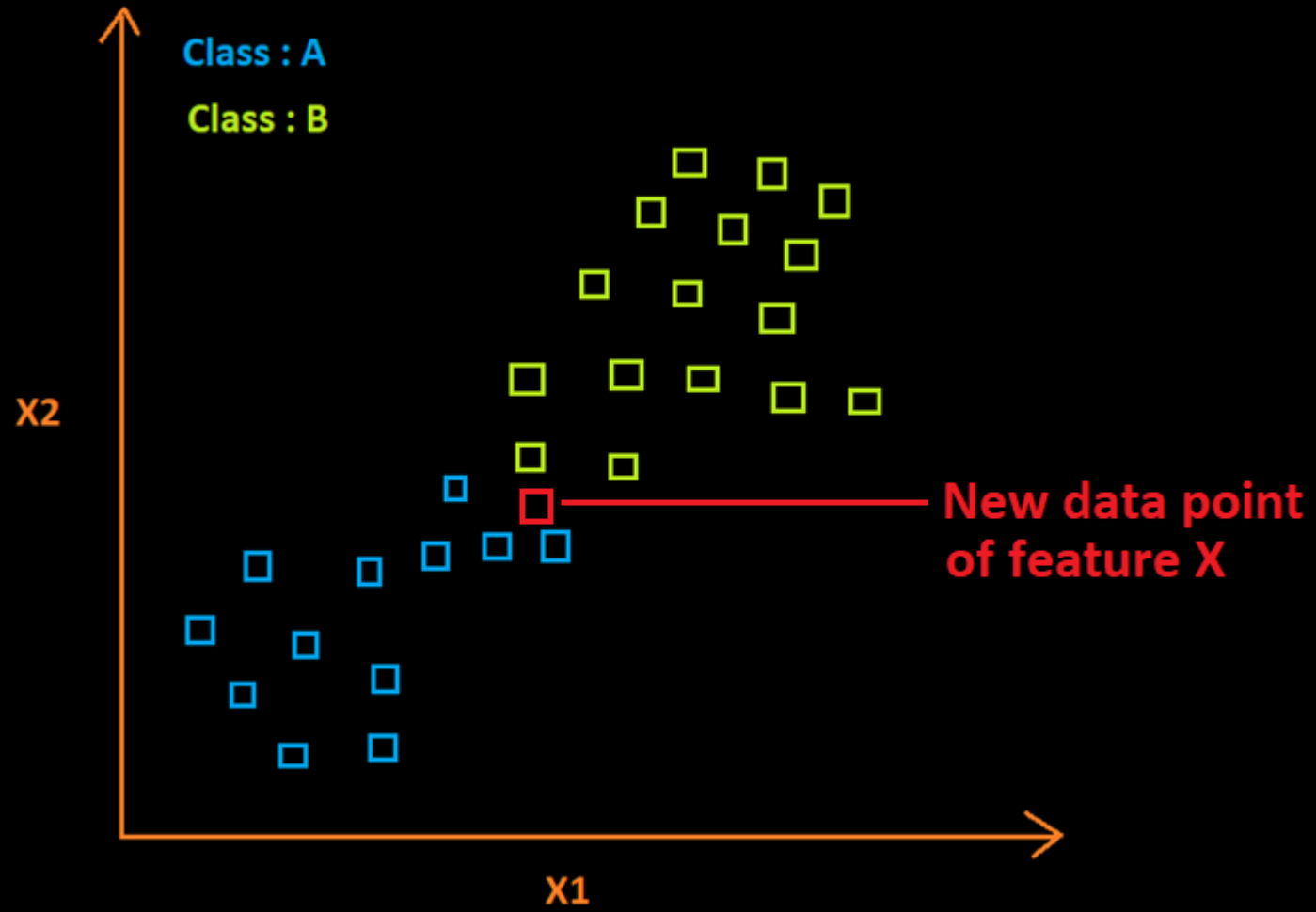
Sample for class B : $\mathbf{X}_2 = (x_1, x_2) = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$

x1	4	2	2	3	4	9	6	9	8	10
x2	2	4	3	6	4	10	8	5	7	8
LDA	4.46	3.48	3.06	5.2	5.3	12.35	8.8	10.2	10.19	12.42

LDA Visualization and class distribution



Naive Baye's Classification



Task

Given a new data point of feature X , find the class for new data point using Naïve Baye's Classifier.

Decision Rule

If

$$P(classA/X) > P(classB/X)$$

then new data point belongs to class A, otherwise belongs to class B

Bayes' Theorem

$$P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

This equation is called Baye's Theorem

Where,

$P(A)$ = Prior probability of event A

$P(A/B)$ = Posterior probability of event A

$P(B/A)$ = Likelihood or Class conditional probability

$P(B)$ = Evidence

Concept of Likelihood

Let

X be the event that we get an X-ray of a person.

Y be the event that a person has a cancer.

$P(Y)$: Probability of having a cancer. It is the prior probability of number of cancer cases coming up. Suppose $P(Y) = 0.1$, it means that 10% people in this region have a cancer.

$P(Y/X)$ is the posterior probability which we are interested in. It is the conditional probability that is, given an X-ray image, what is the probability that person has cancer.

$P(X/Y)$ is the likelihood function which states the class conditional probabilities, i.e., how usually an X-ray of a person having cancer looks like (This is the data which we observe from previous history of all cancer patients in the region).

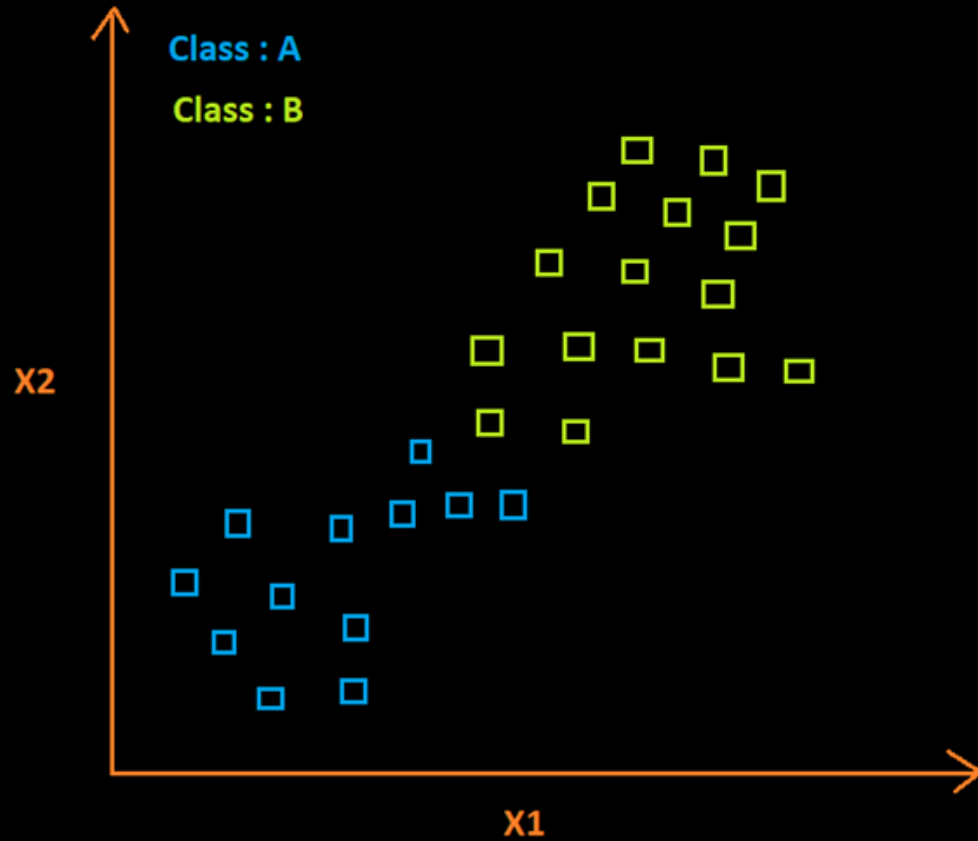
Steps to achieve Task

$$P(classA/X) \quad \text{Vs} \quad P(classB/X)$$

$$\frac{P(X/classA)P(classA)}{P(X)} \quad \text{Vs} \quad \frac{P(X/classB)P(classB)}{P(X)}$$

$$\frac{P(X/classA)P(classA)}{\cancel{P(X)}} \quad \text{Vs} \quad \frac{P(X/classB)P(classB)}{\cancel{P(X)}}$$

$$P(X/classA)P(classA) \quad \text{Vs} \quad P(X/classB)P(classB)$$

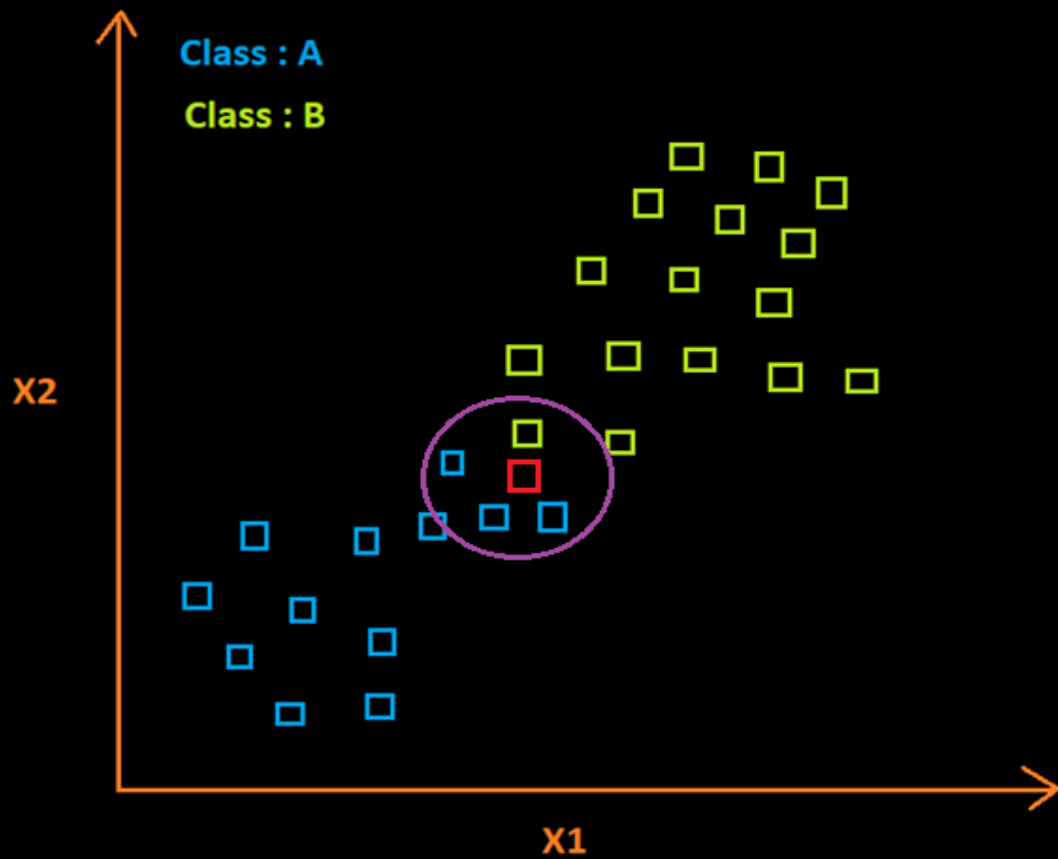


$$P(classA) = \frac{\text{Number of samples in classA}}{\text{Total samples in data}}$$

$$P(classA) = \frac{12}{28}$$

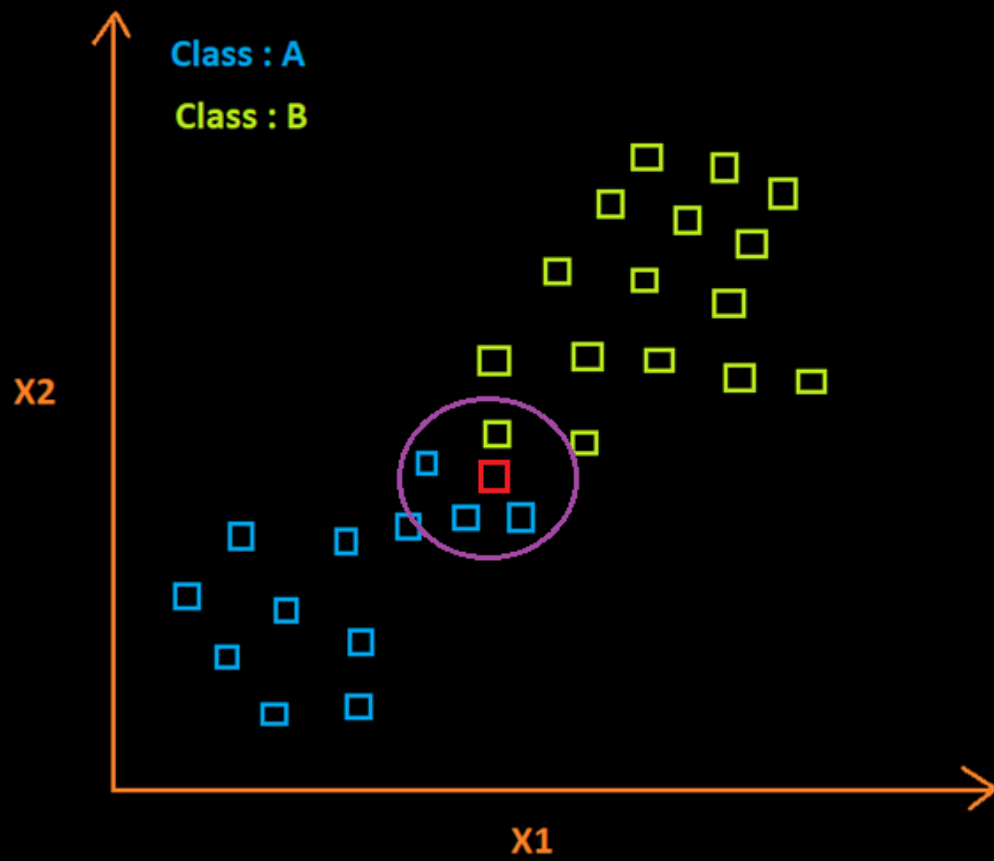
$$P(classB) = \frac{\text{Number of samples in classB}}{\text{Total samples in data}}$$

$$P(classB) = \frac{16}{28}$$



$$P(X/classA) = \frac{\text{Number of similar observations with classA}}{\text{Total samples in classA}}$$

$$P(X/classA) = \frac{3}{12}$$



$$P(X/classB) = \frac{\text{Number of similar observations with classB}}{\text{Total samples in classB}}$$

$$P(X/classB) = \frac{1}{16}$$

Finally,

$$P(\text{class}A/X) = P(X/\text{class}A)P(\text{class}A)$$

$$P(\text{class}A/X) = \left(\frac{3}{12}\right) \times \left(\frac{12}{28}\right) = \frac{3}{28}$$

$$P(\text{class}B/X) = P(X/\text{class}B)P(\text{class}B)$$

$$P(\text{class}B/X) = \left(\frac{1}{16}\right) \times \left(\frac{16}{28}\right) = \frac{1}{28}$$

Decision

Since

$$P(classA/X) > P(classB/X)$$

Therefore, the new data point belongs to class A

